

Formulating a BRICS-5 Currency Basket: An Experiment

DAVID CHI ZHANG

Summary

- This note develops a currency basket comprising the Brazilian Real (BRL), Russian Ruble (RUB), Indian Rupee (INR), Chinese Yuan (CNY, or Renminbi), and South African Rand (ZAR).
- The basket weights are modelled on the International Monetary Fund's Special Drawing Rights (SDR) basket and incorporate international trade and financial variables.
 - Several variables used to calculate SDR basket weights can be replicated for a BRICS-5 basket: exports, outstanding international debt, and FX turnover.
 - Other variables are unavailable and are therefore replaced with proxies.
 - Official reserves denominated in BRICS currencies are proxied by foreign ownership of BRICS countries' local-currency government debt.
 - International banking liabilities denominated in BRICS currencies are proxied by the cross-border local-currency liabilities of resident banks in BRICS countries.
 - The relative international presence of BRICS currencies is consistent across these variables.
- Applying the SDR weighting methodology, the calculated basket is dominated by the Chinese Yuan (nearly 70%). The Indian Rupee ranks a distant second (around 14%), followed by the Brazilian Real (nearly 10%), the Russian Ruble (just over 5%), and the South African Rand (nearly 5%).
- The BRICS-5 Basket exchange rate against the USD broadly tracks BRICS-5 currency movements from 2020 to 2026.
 - Daily changes in the Basket exchange rate against the USD are symmetrically distributed around a median of 0%.
 - The Basket exchange rate against the USD exhibits volatility similar to that of CNY and INR and is more stable than BRL, RUB, and ZAR.

Part I: Modelling a BRICS Currency Basket Based on the IMF SDR

How Are SDR Weights Determined?

According to the [Review of the Method of Valuation of the SDR](#) (IMF, 2022):

- "The relative share of a currency in the basket is determined on the basis of its relative share in exports and a number of financial variables."

Category	Indicator	Weight	Source
Exports	A country's share of exports of goods and services (nominal terms, in SDR, excluding intra-EU trade)	0.5	IMF and European Central Bank
Financial	A currency's share of central bank official reserves	$0.5 \times 1/3$	IMF Currency Composition of Official Foreign Exchange Reserves survey
Financial	A currency's share of foreign exchange turnover	$0.5 \times 1/3$	BIS Triennial Central Bank Survey of Foreign Exchange and Over-the-counter Derivatives Markets
Financial	A currency's share of total international banking liabilities and outstanding international debt securities	$0.5 \times 1/3$	BIS Locational Banking Statistics and BIS Debt Securities Statistics

The SDR Weight Formula

- In formula form, the weight ω_i of currency i in the IMF SDR basket is given by:

$$\omega_i = \left(0.5 \times \frac{X_i}{\sum_j X_j}\right) + \left(\frac{1}{6} \times \frac{R_i}{\sum_j R_j}\right) + \left(\frac{1}{6} \times \frac{FX_i}{\sum_j FX_j}\right) + \left(\frac{1}{6} \times \frac{FA_i}{\sum_j FA_j}\right)$$

- X_i — exports of goods and services of the country issuing currency i
 - R_i — official reserve in currency i
 - FX_i — foreign exchange turnover denominated in currency i
 - FA_i — sum of international banking liabilities and international debt securities denominated in currency i
- • The sums cover all five currencies in the SDR basket. The export component has a weight of 0.5, while reserves, FX turnover, and financial assets each have a weight of 1/6.

Replication Feasibility

- Exports
 - **Data are available** for all five BRICS founding members from the World Bank World Development Indicators database.
 - The measure could be broadened to include both exports and imports, better capturing a country's role in international trade.
- Official Reserves
 - **Data are not available** separately for the currencies of Brazil, Russia, India, and South Africa, which are grouped under "Others" in the IMF COFER survey.
 - Proxy: foreign ownership of a country's local-currency government debt, based on IIF data.
 - Central bank FX reserves are generally held in a currency's most liquid assets, such as bank deposits and treasury securities. Therefore, greater foreign ownership of a country's local-currency government debt may indicate a larger potential share of that currency in international official reserves.
- Foreign Exchange Turnover
 - **Data are available** for all five BRICS founding members from the BIS Triennial Central Bank Survey.
- International Banking Liabilities
 - **Data are not available** separately for the currencies of Brazil, Russia, India, and South Africa.
 - Proxy: a country's cross-border claims and liabilities denominated in its domestic currency.
 - For currency X, considering only the cross-border local-currency liabilities of banks resident in the issuing country excludes deposits in offshore markets. Because cross-border claims in local currency capture at least some offshore local-currency assets, adding them to cross-border liabilities partially addresses this underestimation.
- International Debt Securities Outstanding
 - **Data are available** for all five BRICS founding members from the BIS International Debt Securities database.

Part II: Historical Presence of BRICS-5 Countries and Currencies in Global Trade and Finance

Exports

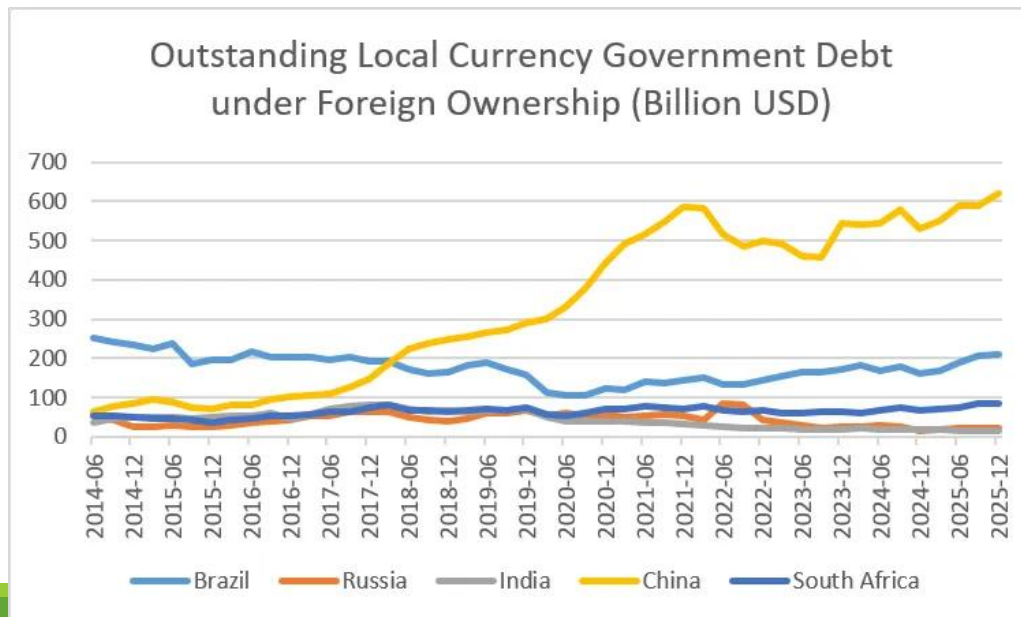
- China has had the largest share of global exports among the BRICS-5 countries over the past two decades.
 - Its share has increased more slowly in recent years and stood at 11.8% in 2025.
- India's share of global exports has trended upward over the past two decades.
 - It now ranks second in the group, at 2.5% in 2025.
- Russia had the second-largest share of global exports among the BRICS-5 countries before 2014, but its share has since trended downward, reaching 1.3% in 2025.
- Brazil's and South Africa's shares were relatively stable over the past two decades, at 1.2% and 0.4% in 2025, respectively.
- Adding imports changes the BRICS-5 countries' relative weights only marginally: China's share falls by 2 percentage points, while India's rises by a similar amount.



Implied BRICS-5 Currency Basket Weights Based on International Trade		
Trade	Exports	Exports + Imports
Brazil	6.8%	7.2%
Russian Federation	7.8%	7.7%
India	14.7%	16.4%
China	68.5%	66.3%
South Africa	2.2%	2.3%

Foreign Ownership of Local-Currency Government Debt

- USD values are calculated by multiplying each country's outstanding local-currency government debt by the corresponding foreign-ownership share; both figures are sourced from the International Institute of Finance (IIF).
- China overtook Brazil in 2018 to become the largest BRICS country by foreign ownership of local-currency government debt.
 - Foreign holdings of China's local-currency government debt declined from 2021 to 2023, then recovered to USD 621 billion at the end of 2025.
- Foreign holdings of Brazil's local-currency government debt declined from 2014 to 2020 before rising moderately to USD 209 billion at the end of 2025, the second-highest level among the BRICS-5.
- Foreign holdings of South Africa's local-currency government debt have risen gradually, reaching USD 84 billion at the end of 2025.
- India and Russia have relatively low levels of foreign holdings in local-currency government debt, at USD 23 billion and USD 16 billion, respectively, at the end of 2025.

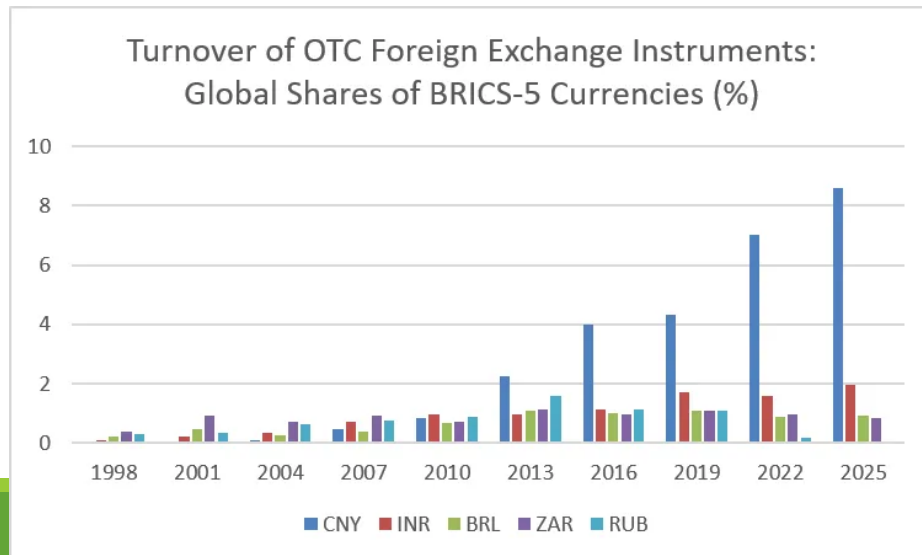


Implied BRICS-5 Currency Basket Weights Based on Foreign-Held Local-Currency Government Debt

BRL	21.9%
RUB	2.4%
INR	1.7%
CNY	65.2%
ZAR	8.8%

Foreign Exchange Turnover

- The BIS's triennial central bank survey provides a currency breakdown of foreign exchange turnover.
- The Chinese Yuan has the largest share of FX trading among BRICS-5 currencies and has grown rapidly, reaching 8.6% in 2025.
- The Indian Rupee's share of FX trading has also increased in recent years and is now the second-highest among BRICS currencies, at 1.95% in 2025.
- The Brazilian Real's global share rose during the first decade of the 21st century but declined in recent years, reaching 0.94% in 2025.
- The South African Rand led BRICS-5 currencies in FX turnover in the early 2000s, but its share declined over the past decade to 0.86% in 2025.
- The Russian Ruble had the second-largest share among BRICS currencies through 2016. Since 2022, however, its share has fallen sharply, reaching just 0.04% in 2025.

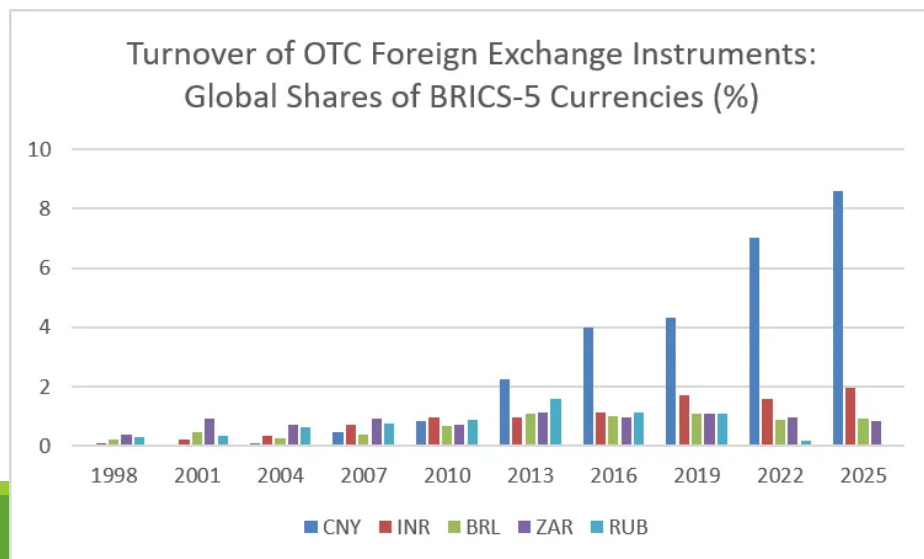


Implied BRICS-5 Currency Basket Weights Based on Foreign Exchange Turnover

BRL	7.6%
RUB	0.3%
INR	15.7%
CNY	69.4%
ZAR	6.9%

International Debt Outstanding

- The BIS International Debt Securities Statistics provides data on outstanding international debt for several currencies, including the BRICS-5. Under the BIS definition, international debt comprises debt securities "issued outside the local market of the country where the borrower resides."
- CNY-denominated international debt accounts for 0.94% of the global total, the highest share among BRICS-5 currencies.
 - The CNY share declined from 2015 to 2022 amid FX reform, trade tensions, and COVID-19, but regained momentum from 2022 to Q1 2026, during which the outstanding amount nearly doubled.
- BRL- and ZAR-denominated debt once led the BRICS-5, but both shares declined over the past decade, reaching 0.14% and 0.09%, respectively, at the end of Q1 2026.
- The INR share rose from 2015 to 2018, eased from 2019 to 2023, and then began rising again, reaching 0.06% in Q1 2026.
- The RUB share has declined since 2014, falling to 0.02% in Q1 2026, the lowest among BRICS-5 currencies.

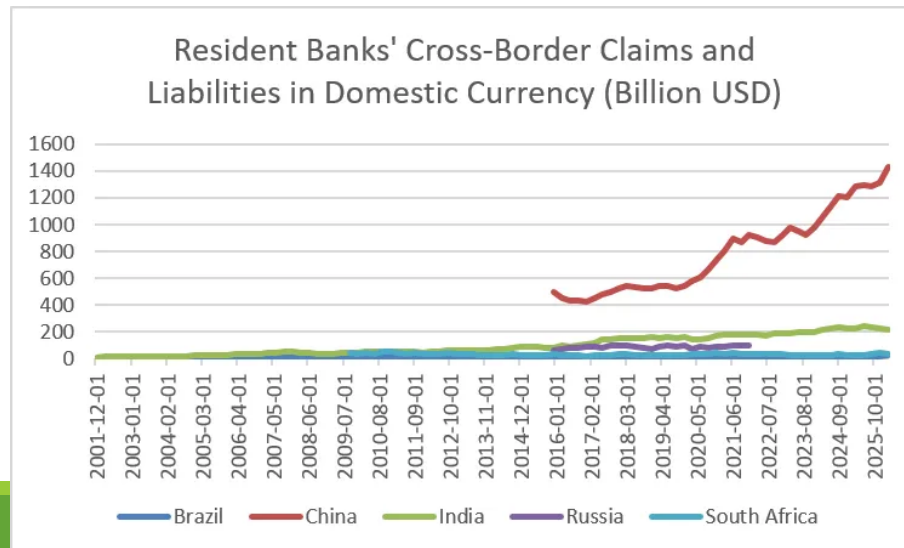


Implied BRICS-5 Currency Basket Weights Based on Outstanding International Debt

BRL	11.1%
RUB	1.3%
INR	5.0%
CNY	75.2%
ZAR	7.5%

Resident Banks' Cross-Border Claims and Liabilities in Local Currencies

- At the end of Q1 2026, banks resident in China had the equivalent of USD 1.4 trillion in cross-border claims and liabilities denominated in the national currency (CNY), the highest level among BRICS countries. The amount has risen steadily.
- Banks resident in India rank second among the BRICS countries, with the equivalent of USD 214 billion in claims and liabilities, also on an upward trend in recent years.
- Banks resident in Russia ranked third among the BRICS countries from 2016 to 2021 but stopped publishing the data in 2022. For calculation purposes, the amount is assumed to remain unchanged in subsequent years.
- Banks resident in South Africa and Brazil have relatively low cross-border positions in their national currencies, at the equivalent of USD 39 billion and USD 24 billion, respectively, in Q1 2026.
- Excluding cross-border claims and using only liabilities would reduce China's weight by about 8 percentage points, while increasing the weights of India and Russia by a few percentage points.



Implied BRICS-5 Currency Basket Weights Based on Resident Banks' Cross-Border Local-Currency Positions

	Claims + Liabilities	Liabilities
Brazil	1.3%	0.6%
China	79.1%	71.4%
India	11.9%	19.0%
Russia	5.5%	6.7%
South Africa	2.2%	2.3%

Summary of Implied Weights Across Variables

- The Chinese Yuan has the strongest international presence among BRICS currencies, with implied weights ranging from 65% to 80% in the BRICS-5 Basket.
- The Indian Rupee ranks second by implied weight in trade, FX turnover, and cross-border banking, despite its smaller presence in international debt and foreign ownership of local-currency government debt.
- The Brazilian Real ranks relatively high in international debt and foreign ownership of local-currency government debt.
- The Russian Ruble and South African Rand once had a greater international presence but now rank below the other three BRICS currencies.

Category	FX Trading	International Debt	Cross-Border Local-Currency Banking	Cross-Border Local-Currency Banking	Government Debt	Trade	Trade
Indicator	A. FX Turnover	B. International Debt Statistics	C. Liabilities + Claims	D. Liabilities	E. Foreign Ownership of Local Currency Government Debt	F. Exports	G. Exports + Imports
BRL	7.6%	11.1%	1.3%	0.6%	21.9%	6.8%	7.2%
RUB	0.3%	1.3%	5.5%	6.7%	2.4%	7.8%	7.7%
INR	15.7%	5.0%	11.9%	19.0%	1.7%	14.7%	16.4%
CNY	69.4%	75.2%	79.1%	71.4%	65.2%	68.5%	66.3%
ZAR	6.9%	7.5%	2.2%	2.3%	8.8%	2.2%	2.3%

Two Weighting Schemes for the BRICS-5 Currency Basket

- Two baskets

- Basket 1: $W_1 = 0.5 \times \left(\frac{1}{4}A + \frac{1}{4}B + \frac{1}{4}C + \frac{1}{4}E\right) + 0.5 \times F$

- Basket 2: $W_2 = 0.5 \times \left(\frac{1}{4}A + \frac{1}{4}B + \frac{1}{4}D + \frac{1}{4}E\right) + 0.5 \times G$

- Basket 2 is less concentrated than Basket 1, although the differences are marginal.

- Basket 2 is used as the default in the analysis that follows.

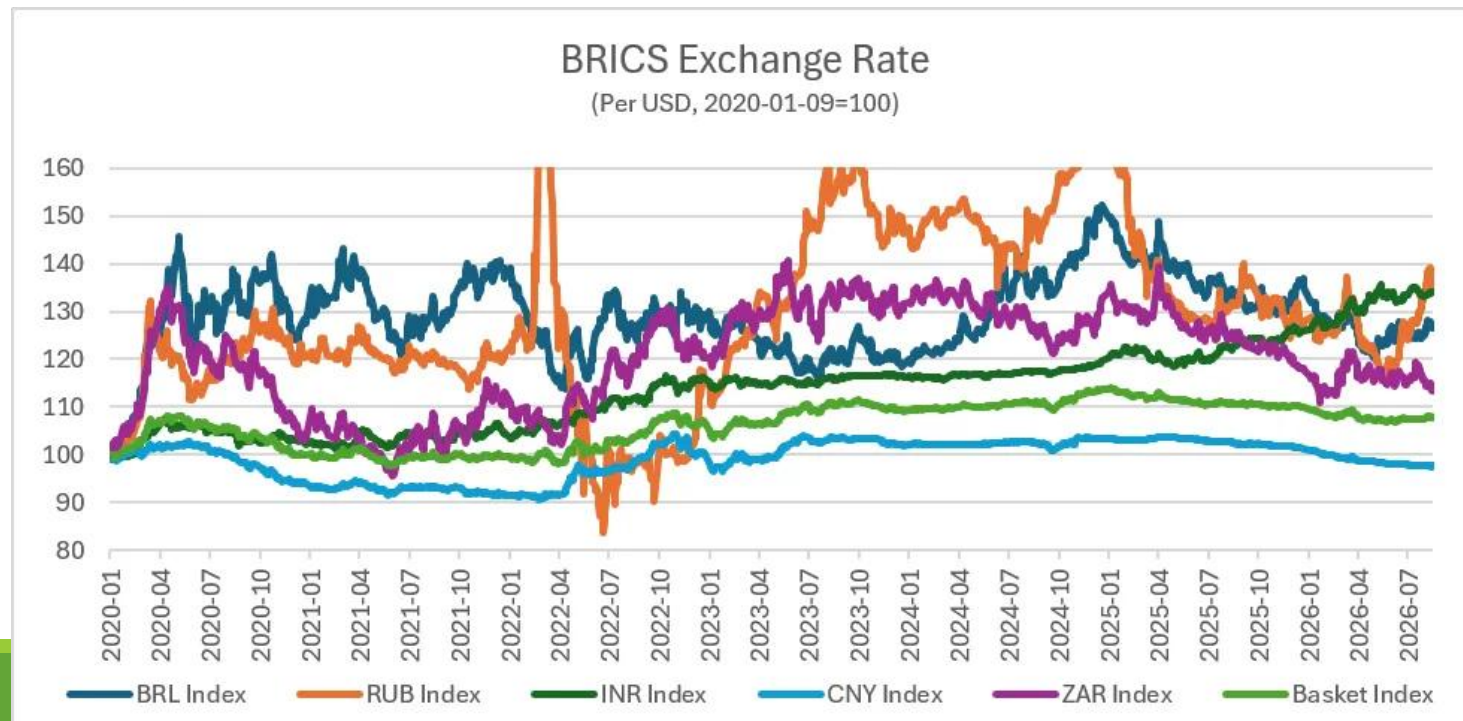
- Setting the basket's face value at USD 10 on August 24, 2026, using that day's exchange rates, yields **BRL 4.5, RUB 43.2, INR 128.1, CNY 46.3, and ZAR 7.0.**

Category	FX Trading	International Debt	Cross-Border Local-Currency Banking	Cross-Border Local-Currency Banking	Government Debt	Trade	Trade
Indicator	A. FX Turnover	B. International Debt Statistics	C. Liabilities + Claims	D. Liabilities	E. Foreign Ownership of Local Currency Government Debt	F. Exports	G. Exports + Imports
BRL	7.6%	11.1%	1.3%	0.6%	21.9%	6.8%	7.2%
RUB	0.3%	1.3%	5.5%	6.7%	2.4%	7.8%	7.7%
INR	15.7%	5.0%	11.9%	19.0%	1.7%	14.7%	16.4%
CNY	69.4%	75.2%	79.1%	71.4%	65.2%	68.5%	66.3%
ZAR	6.9%	7.5%	2.2%	2.3%	8.8%	2.2%	2.3%

Part III: Evaluating the Basket

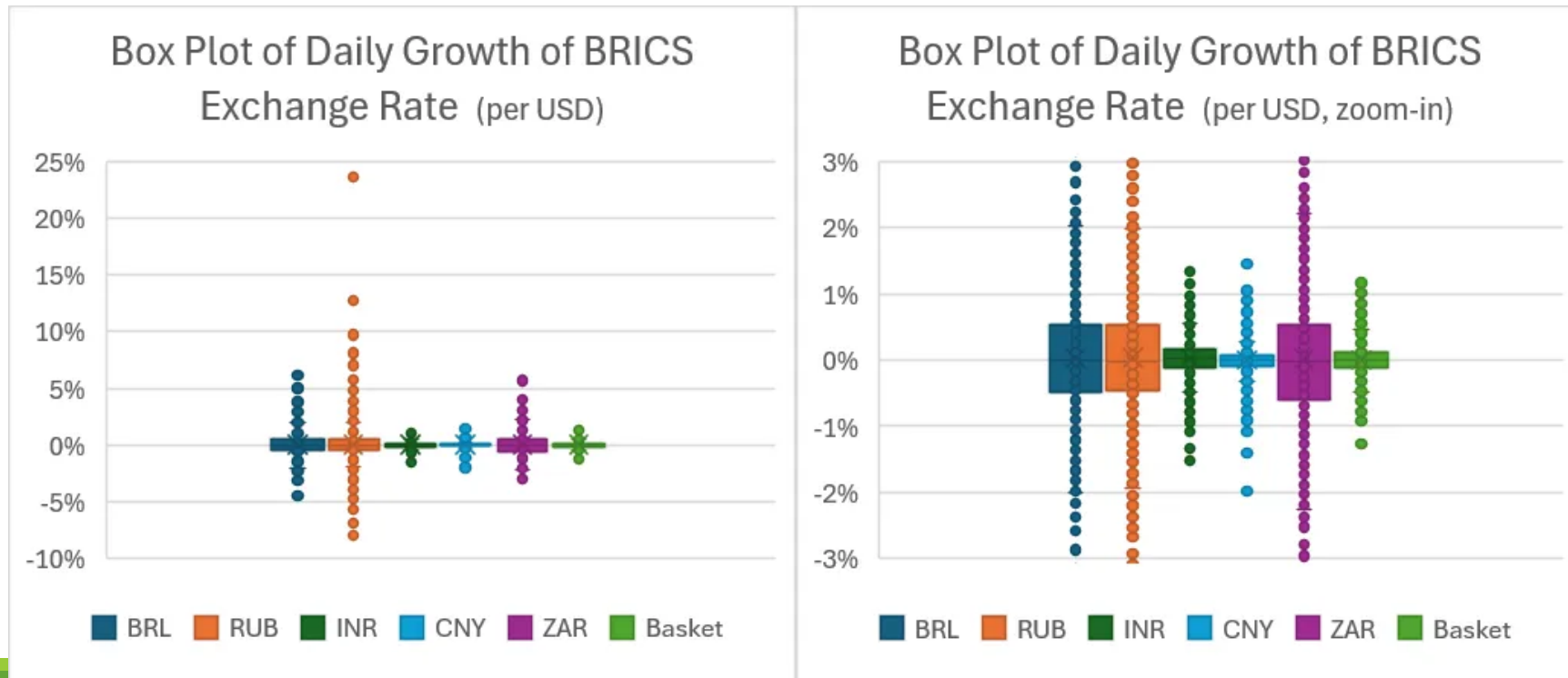
BRICS-5 Exchange Rates Against the USD

- The chart indexes the exchange rate of each BRICS-5 currency and the Basket (per USD) to January 2020.
- Except for Ruble movements in 2022-2023, which were largely driven by geopolitical factors, BRICS-5 exchange rates against the US dollar broadly followed similar trends. The Ruble, Rand, and Real varied more than the Rupee and Renminbi.
- The Basket exchange rate against the US dollar varies less than those of the individual BRICS-5 currencies.
 - Because BRICS-5 currencies generally move in the same direction against the USD, the Basket captures their broad trend, although its movements are more heavily influenced by the Renminbi.



Distribution of Daily Percentage Changes in Exchange Rates Against the USD

- The distributions of all BRICS-5 currencies and the Basket are centered around a mean of 0%.
- The distributions of the Basket, CNY, and INR are symmetric, while those of BRL, RUB, and ZAR are skewed toward depreciation.
- The Basket's interquartile range (IQR) of 0.236 percentage points is larger than CNY's (0.150), comparable to INR's (0.260), and smaller than those of BRL (1.011), RUB (0.986), and ZAR (1.125).



Volatility of Daily Percentage Changes in Exchange Rates Against the USD

- The Basket's exchange-rate volatility against the USD is comparable to that of CNY and INR.
 - The five-year averages of monthly standard deviations for the Basket, CNY, and INR are 0.22%, 0.19%, and 0.27%, respectively.
- The Basket is much less volatile than BRL, RUB, and ZAR.
 - For BRL, RUB, and ZAR, the corresponding averages are 0.93%, 1.16%, and 0.93%, respectively.

